What Does Repairing Choice Inconsistency Actually Buy? A Budget-Set Diagnosis

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Abstract

Repairing an AI agent's incoherent preferences is not new: inference-time transitivity repairs, isotonic projections onto preference-defined cones, and training-time rationality penalties have all been proposed, several with reported downstream gains. None controls for displacement magnitude: no result establishes whether restoring coherence itself, not mere displacement toward an interior point, is what any reported gain comes from. We build that control. Given an LLM agent's own choices over budget sets, we compute the minimal MILP perturbation restoring GARP-consistency, then compare it against a GARP-blind null operator spending the identical displacement budget shrinking every bundle toward a fixed center. Under the paper's original exogenous payoff, the null outperforms the real repair by more than 2x (mean payoff gain 0.0220 vs. 0.0091, Wilcoxon $p = 3.98 \times 10^{-10}$, winning 81% of traces). A second, independently designed payoff built to remove the first one's exploitable fixed target reproduces the same verdict (null 0.0217 vs. real 0.0072, $p = 1.24 \times 10^{-5}$, 70.8% win rate); we decline a third as payoff-shopping. Trace extremity predicts the null's advantage in both experiments. The paper's standing contributions are instead a capacity-deconfounded identification strategy for the coherence–competence question, which remains open, and a discard-selection instrument-validity finding.

1 Introduction

An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will frequently violate the Generalized Axiom of Revealed Preference (GARP; Afriat, 1967, Varian, 1982): it prefers bundle x_1 to x_2 at one budget line and the reverse at another, in a way no monotone utility function can rationalize. The same budget-allocation instrument is the standard paradigm for testing GARP experimentally on human subjects [Andreoni and Miller, 2002, Choi et al., 2007, 2014], and a growing literature applies it to GPT-family models directly, without a repair step, to ask whether they satisfy classical rationality axioms at all [Chen et al., 2023]. A separate, growing literature repairs exactly this kind of inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal scores onto monotone cones, penalizing incoherence during training — and several of these systems report that the repaired outputs are downstream-better than the raw ones. What none of them establishes is a *dose*: how much better, as a function of how much repair was needed, isolated from the capacity and training confounds that come from comparing a richer model class to a poorer one.

We study this question in a setting where it can be answered cleanly. An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a single mixed-integer linear program, the minimal quantity perturbation that would make the agent's own realized choice

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sequence GARP-consistent — a projection applied *post hoc* to a frozen agent, so nothing about its weights, training, or decoding changes across doses. The size of that perturbation is a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and income alone — never of the agent's own choices — so the outcome measure is exogenous to the preference data by construction, not a restatement of coherence. Tracing dose against Δpayoff gives the relationship this paper reports.

That relationship turns out not to isolate what it was built to isolate. A control absent from every published axiom-enforcement result we are aware of — an operator that knows nothing about GARP but spends the identical displacement budget shrinking every bundle toward a fixed interior point — outperforms the real GARP-restoring repair on the exogenous payoff, significantly, at both model scales, under two independently designed payoffs (§5.1). This is the paper's central finding: restoring rationality, as originally proposed, does not buy more exogenous payoff than mere displacement toward the interior of the budget line does on its own. We report this as a negative result reached twice by construction, not as a null we happened to observe once.

We do not claim any of the individual pieces are new; §2 concedes exactly what is not. What has not been done is the conjunction: an agent's own realized choices, a graded coherence-indexed dose, and an outcome that is not itself a preference judgment, run together so that the identification is clean. We run this at a model scale where a pilot study found measurable coherence headroom (1.5B parameters), and repeat the identical pipeline at a scale with almost none (3B parameters) as an identification control on the method itself, not as a second point on a shared dose curve across scale.

Two further results were not predicted by the design. The reciprocal-price framing manipulation intended to induce coherence variation at 1.5B produced, in a pilot run, a large apparent effect that disappears once discard-selection is corrected with a capped retry protocol: the pilot's naive estimate and the corrected main-experiment estimate are both statistically indistinguishable from zero, and the two should not be read as a sign reversal of a real effect. We report the discard rate itself, not the CCEI point estimate, as the more reliable signal that the manipulation disrupted the agent (our claim C3). Wang et al. [2025a] already recommend reporting invalid-response proportions as a checklist item; our contribution is a quantitative demonstration of how much a naive discard rule can distort a measured effect at this specific scale, broken down by attempt outcome in §5.2, not the recommendation itself. Separately, splitting single-turn elicitation into separate sequential calls at 1.5B produces a large GARP-pass-rate collapse, moving in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same domain and model family at a larger scale, with no discard confound on either arm.

Contributions. (1) An application — not a novel method — of Demuynck and Rehbeck [2023]'s minimal-quantity-error MILP projection to an agent's own realized choices, independently verified on every output rather than trusted from solver status alone. (2) A closed-form exogenous payoff, fixed before any data collection, audited adversarially for leakage and for the mechanical confound that larger violations simply have more room to improve. (3) A distance-matched, GARP-blind null-operator control — absent from every published axiom-enforcement result we are aware of — run against two independently designed exogenous payoffs, showing that the measured raw dose–response relationship is not evidence that GARP-restoration specifically, rather than mere displacement toward the interior, is what buys the exogenous payoff gain (§5.1). (4) An instrument-validity finding (discard-selection bias under a disruptive manipulation), corrected for by a stated retry protocol and reported per cell rather than pooled away. (5) A single-turn-vs-multi-turn elicitation-format manipulation that produces a large GARP-pass-rate collapse at 1.5B, in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same budget-allocation domain and the same model family (Qwen2.5) at a larger scale.

2 Related Work

Repairing an AI system's incoherent preferences is not new; Table 1 positions nine published systems against the four criteria that jointly define our claim: an agent's own choices (not a judgment about third-party items), an exogenous payoff (no preference judgment enters it), a graded rather than binary dose, and a formal minimum-distance projection. None occupies all four.

Table 1: Where prior repair systems sit. *Own choices*: the repaired object is a choice the agent made for itself from a budget set, not a judgment about third-party items. *Exogenous payoff*: an outcome into which no preference judgment enters.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	**yes**
Rationality layer [Chadwick et al., 2025]	no	no	no	**yes**
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	**yes**	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	**yes**	no	no	no
GARP-EFM [Aguiar and Kashaev, 2026]	no	**yes**	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	**yes**	no
Control vectors [Cook et al., 2026]	**yes**	no	**yes**	no
This paper	**yes**	**yes**	**yes**	**yes**

POISE [Wang et al., 2026] is the sharpest vocabulary collision: pool-adjacent-violators projection onto a closed convex chain-monotone cone, with a proved Pythagorean guarantee the edit lands weakly closer to a posited ground truth. We concede the minimum-distance priority unqualified: projection onto a closed convex set is non-expansive in L_2, licensing that guarantee; the GARP-consistent set is a union of polyhedra, hence not convex, so no analogue transfers here (§3.2). POISE also projects a *teacher's* offline labels, not an agent's own choices. Three further inference-time repairs, each on a different object: Chadwick et al. [2025] (synthetic election data, no downstream comparison), TrustJudge [Wang et al., 2025b] (third-party judgments, no degree parameter), CONSISTRE [Sun, 2026] (relation-extraction triples, no Afriat machinery). LLM-RankFusion [Zeng et al., 2024] scores its rankings against real TREC relevance labels — the only one of these four with a genuinely exogenous payoff — but explicitly declines a minimum-distance rank-aggregation method, naming Kemeny aggregation and rejecting it as NP-hard in favor of simple Borda-count voting across rankers. One training-time system alters the agent itself: Buchanan and Foster [2026] (IIA-invariance loss, no downstream evaluation). A further training-time system is not an LLM agent at all: Aguiar and Kashaev [2026] fine-tune Chronos-2, a time-series forecaster, on synthetic utility-maximizing-agent data to predict human consumers' own choices, with a downstream evaluation (17–31% bundle-prediction error reduction) neither of the other two above reports.

A parallel line varies the transitivity of the *preference model* rather than an agent's choices: GPM [Zhang et al., 2025] and HRC/DSPPO [Huang et al., 2026] both make the cycle-tolerant arm a strict superset model class, confounding coherence with capacity by construction, and on GPM's own length-controlled metric that arm *loses* in 18 of 24 cells. HRC/DSPPO already publishes its own inverted-U dose–response curve over a training-time schedule weight — we say so before claiming anything about grading — but the dose weights a learned third-party proxy during training, not an agent's own realized choices, and the outcome is an LLM-judge score throughout, where ours acts post hoc and is exogenous. Our closest theoretical neighbour, Andrews [2026], proposes $1 - \mathrm{CCEI}$ as a training penalty and states plainly, repeatedly, that coherence is *not* sufficient for good behaviour — argued a priori, never measured, never asking whether *imposing* coherence helps or hurts; ours is the empirical counterpart to a proposal that has circulated unrun.

The sharpest objection is psychometric, and it is in *PNAS*. Nitsch et al. [2022] find CCEI/Houtman–Maks [Houtman and Maks, 1985] reliability never reaches acceptable levels across eight datasets, and that participants given the chance to revise their own inconsistent choices saw mean CCEI *fall*. That revision arm is not a repair operator (a random subset of choices, no consistency objective, no guarantee of fewer violations) — ours attains rationalizability by construction and verifies it independently, the same distinction that disposes of Yamin et al. [2026a]'s isotonic-calibration repair (worse in 14 of 16 cells, projecting onto a different set than the one it is graded on). A companion paper by an overlapping author team fits revealed-preference-derived cost functions to LLM decisions to diagnose alignment and steerability rather than to repair anything [Yamin et al., 2026b] — the same GARP/Afriat machinery used here for measurement rather than projection. The reliability finding is diagnosed by its own authors as a between-subject-variance problem whose prescribed remedy — a between-groups design — is this paper's design; the concession we do not shed is that three independently published axiom-enforcement results (adding Zhu et al. [2025]'s) all point the same adverse way, and a null result here would be a fourth confirmation rather than a discovery, which is why we carry a distance-matched null-operator control neither published negative had.

The closest neighbour on the economics side is invisible to any arXiv sweep. Cook et al. [2026] steer economic choices toward payoff-maximising behaviour via personas, prompt masking, and control vectors — occupying two of our three legs without Afriat machinery or a minimum-perturbation objective; their finding that reframing moves models while persona prompting does not corroborates our manipulation choice, anchored in Wang et al. [2025a]. Wang et al. find *format*, not persona or temperature, moves the Afriat index in the same budget-allocation domain we study, on Qwen2.5-7B-Instruct among other models — the same model family as our 1.5B, at a larger scale: collapsing their multi-turn baseline to single-turn drops CCEI by up to 0.241. Our arm runs the opposite manipulation (single-turn baseline split into multi-turn) and finds the opposite direction: GARP pass rate collapses from 0.40 to 0.10 ($p = 0.0073$; survives Benjamini–Hochberg but not Holm correction across our full test family) with zero discards. This is not a replication: in the same domain and model family, which format is more coherence-degrading reverses between their 7B model and our 1.5B model. We report the disagreement rather than the confirmation we expected.

3 Method

3.1 The projection: Demuynck & Rehbeck's (2023) minimal-quantity-error MILP, applied and independently verified

This section applies, rather than proposes, a projection method. The MILP formulation below is Demuynck and Rehbeck [2023]'s multiplier-free ordinal characterization of minimal-quantity-error GARP repair, adopted here on an LLM agent's own choice sequence and verified independently on every output; nothing about the formulation itself is new to this paper.

For a sequence of T observations $(p_t, x_t)_{t=1}^{T}$, K goods, GARP holds if there is no sequence of choices such that x_{t_1} is revealed preferred to x_{t_2}, ..., x_{t_n} is revealed preferred to x_{t_1}, with at least one strict revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation $\tilde{x}_t$ that restores GARP-consistency. The naive formulation parameterizes this with Afriat multipliers, which makes the resulting constraints bilinear in the unknowns once $\tilde{x}$ is itself a decision variable — fixing a candidate preference ordering does not remove the bilinearity, since a price-weighted multiplier term remains a product of two unknowns regardless of ordering. We instead use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels $u_t \in [0, 1]$ that carry no multipliers, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. At $T = 25$ this is 600 binaries and 125 continuous variables per trace, solved as a single MILP on the HiGHS backend [Huangfu and Hall, 2018], with no outer search over preference orderings and no alternating scheme.

Three method commitments, each carrying an explicit risk. **Budget exhaustion is imposed as an equality**, $p_t \cdot \tilde{x}_t = I_t$ for every observation (income fixed at 100 throughout this study); this is standard for the design, and it makes the big-M constant in the ordering constraints computable a priori as $\alpha > \max_t I_t$. **A fixed strict-preference margin** $\gamma > 0$, set to $10^{-4} \cdot \min_t I_t$, converts an unattained infimum into an attained minimum: the GARP-consistent set is not closed, so an unregularized projection can read a distance of exactly zero on genuinely violating data. We verified the reported distance is stable across $\gamma \in \{10^{-2}, \ldots, 10^{-6}\}$ before treating it as reportable. **Every returned $\tilde{x}$ is verified independently** via the same combinatorial Warshall-closure [Warshall, 1962] GARP check used everywhere else in this paper, never trusted from solver optimality status alone — all 85 GARP-violating traces' projections in the main experiment were independently re-verified GARP-consistent, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

The objective is L_1: $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform weights $w = 1$. This keeps the program a pure MILP; L_∞ distance is recorded alongside L_1 for every trace but is not separately analyzed in this paper. L_2, the literature's own least-squares index, would require an MIQP solver we do not have configured and is not computed. The *dose* for a trace is its L_1 projection distance $\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$, comparable across sessions without further normalization because income is fixed throughout — the same style of graded, minimum-cost severity measure as Dean and Martin [2016], applied here to an LLM's own choices rather than lab or survey human data and paired with an exogenous behavioral payoff rather than left as a standalone index. A Cobb–Douglas demand share-fitted to each trace's own observed data is computed as a feasibility incumbent and sanity ceiling on the reported distance, recorded only as an upper-bound diagnostic; Appendix A(1) checks in detail that it never reaches the solver and so never influences any reported projection.

3.2 What can be guaranteed, and what cannot: positioning against convex projection

§2 positions this projection against the closest published structural analogue, a proved-optimal projection onto a convex monotone cone [Wang et al., 2026]: because the GARP-consistent set is a union of polyhedra rather than a single convex set, the non-expansiveness that licenses a weakly-closer-to-ground-truth guarantee there has no analogue here. Concretely for this formulation, the prior operator's cone is fixed by an ordering supplied as input, whereas a GARP repair must decide which revealed-preference comparisons to give up — the ordering search is absorbed into the binary comparison indicators of §3.1 rather than eliminated, which is why no distance-minimization guarantee analogous to the convex case is claimed.

3.3 The exogenous payoff

The payoff must not be derived from the agent's own revealed choices — a utility function fit to the agent's choices (such as the projection's own feasibility incumbent above) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences rather than measuring anything external to them. We instead fix, before any model was queried and never re-estimated from any agent's choices, an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5 I_t / p_{t,A}$, $x_{t,B}^* = 0.5 I_t / p_{t,B}$ — a function of prices and income alone, never of the agent's observed or projected bundle. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t)/U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This mirrors — not the specifics, but the structure of — a money-metric utility index, the standard device for welfare comparison over the same GARP/Afriat objects this paper's method already relies on. A genuinely exogenous, real-world payoff of this kind is rare in this literature: Ouyang et al. [2024] score LLM-generated capital-expenditure forecasts against firms' realized future spending and find the relationship between alignment intensity and forecast accuracy is *non-monotonic* — light alignment helps, over-alignment hurts — but without a placebo intervention of matched size, that design cannot separate an alignment-specific effect from a generic effect of any fine-tuning intervention, the same separation our dose–response design is built to make.

We audited this design adversarially, before treating any dose–response result as reportable, for four failure modes: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction secretly aimed at the payoff optimum, and the confound that larger-dose traces simply start worse and have more room to improve. None survived the audit (Appendix A gives every check and number); the raw dose–response statistic we report in §5.1 is not an artifact of any of the four. A fifth failure mode — that this payoff's own concavity could reward mere displacement without any GARP-specific effect — required a new control rather than an audit of the existing design, and is addressed separately in §3.4.

3.4 Two robustness controls against a payoff-geometry confound

The payoff of §3.3 reduces algebraically to $\text{payoff}(s) = 2\sqrt{s(1-s)}$, where s is the expenditure share on good A: concave, with a single interior maximum at $s = 0.5$ identical across every trace. Any operator that moves a bundle toward that one fixed point raises its payoff, whether or not the movement has anything to do with restoring GARP-consistency. To test whether the dose–response relationship of §5.1 reflects GARP-restoration specifically rather than this geometric fact, we construct a **null operator**: for each GARP-violating trace, shrink every observed bundle toward the exogenous optimum x_t^* by a single shared factor $\lambda \in [0, 1]$, chosen so the null's total L_1 displacement exactly matches the real projection's own dose on that trace. The null never calls the GARP check and does not, in general, restore consistency; it is scored by the same payoff function used everywhere else.

Because the first payoff's exploitable property is a single fixed target identical across every trace, we build a second, independently designed payoff that removes it: a per-trace Cobb–Douglas weight

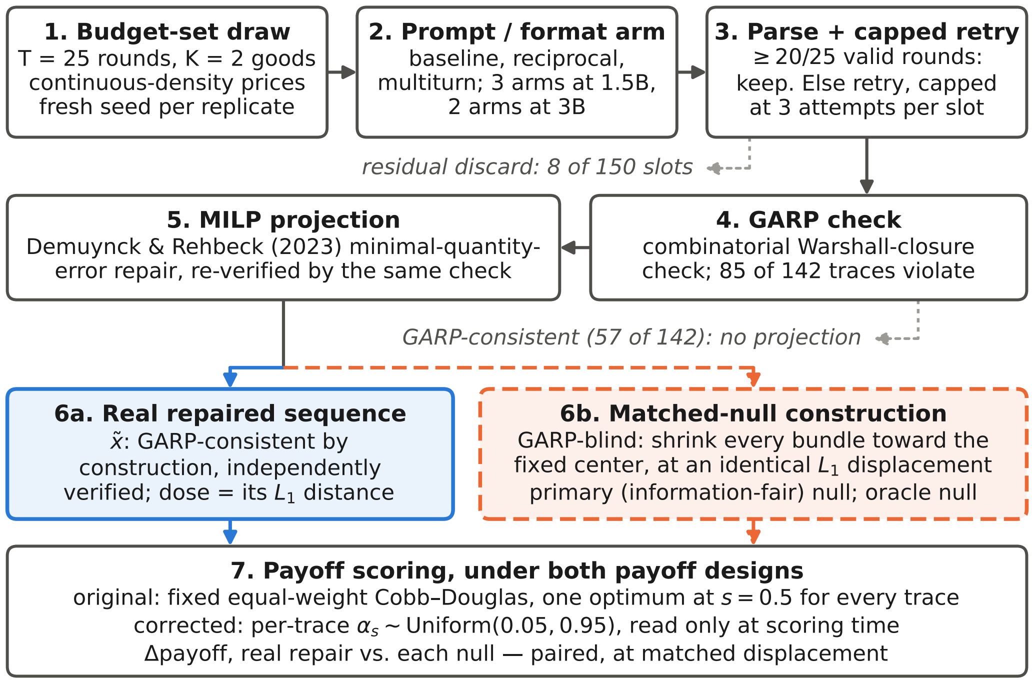

Figure 1: The pipeline, one replicate slot end to end: budget-set draw (1); prompt/format arm (2); parse under the capped retry protocol of §4.3, residual discards excluded (3); combinatorial Warshall-closure GARP check (4); and, for the traces that violate it, the minimal-quantity-error MILP of §3.1 (5), whose L_1 displacement is the dose. Steps 6–7 are the paper's central comparison: the real repaired sequence (6a, solid blue) and the GARP-blind null spending an identical L_1 displacement (6b, dashed orange) are scored by the same step (7) under both payoff designs. Counts are the main experiment's (§5).

$\alpha_s \sim \mathrm{Uniform}(0.05, 0.95)$, drawn independently for each of the 85 traces and for each of $K = 20$ replicate draws, read only at scoring time and never by the projection or the null construction. Under this second payoff we report two null constructions, never pooled: a **primary**, information-fair null (the identical construction above, unaware of α_s) and an **oracle** null (aims directly at the true per-trace optimum $x^*_{\alpha_s}$), reported only as an upper bound since it uses information no real repair algorithm has. Figure 1 summarizes the full pipeline, from the budget-set draw through to the paired scoring of the real and null sequences under both payoff designs.

4 Experimental design

4.1 Models and conditions

Two locally-hosted open-weight models, run entirely on local compute at zero API cost: `qwen2.5:1.5b-instruct` (the headroom model, per a pilot study that found measurable within-condition coherence variation at this scale) and `llama3.2:3b-instruct` (the null-effect control, per the same pilot finding almost no coherence headroom at this scale — mean baseline CCEI 0.99). A third model family was tested for feasibility but excluded after a reproducible multi-model-residency crash under sustained rotation on the local machine; the two-model, model-major design (all sessions for one model complete before the next model loads) is both a wall-clock optimization and, after that observed failure, a correctness requirement. **Compute.** All model inference ran locally via 4-bit-quantized weights on a single consumer CPU-class machine, no GPU and no commercial API calls; the full $N = 30$ design across both models and every condition completed in approximately 2.1 hours wall-clock. Every MILP projection (§3.1) solved in under 5 seconds on the same machine, CPU-only, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

Three conditions at 1.5B, two at 3B. **Baseline**: direct per-token-return framing, single turn, all 25 rounds in one prompt and one response. **Reciprocal**: identical single-turn format with an inverted-price framing manipulation, the strongest manipulation identified in the pilot. **Multiturn** (1.5B only): baseline framing, but each of the 25 rounds is a separate sequential API call carrying the running conversation history, isolating the elicitation-format mechanism from the price-framing mechanism; the arm is motivated by an independently published finding, in the same budget-allocation domain we study, that collapsing multi-turn elicitation to single-turn moves CCEI by up to -0.241 for a same-family model at a larger scale [Wang et al., 2025a], an order of magnitude larger than this paper's own pilot framing effect. Our arm tests the manipulation in the opposite direction (single-turn baseline split into multi-turn), since that is the direction our own baseline format already uses. It is not added at 3B, whose role is a null-effect control on the identification strategy rather than a second scale point requiring the full condition set.

4.2 Budget sets and power

Each of $N = 30$ independent replicates per (model, condition) cell draws its own fresh $T = 25$, $K = 2$ budget set (continuous-density prices, $M, N \sim U[0.1, 1.0]$ i.i.d. rejected unless $\max(M, N) \geq 0.5$, budget exhaustion enforced), seeded independently from every other replicate — unlike a pilot study's single shared set, needed because the main experiment targets independent between-condition replication, not test–retest reliability. Continuous, independently-drawn prices with enforced budget exhaustion also avoid a known pathology: CCEI can read exactly 1.0 on data that violates GARP whenever a comparison lands on exact budget-line equality, an event common under grid-sampled or rounded prices [Andrews, 2026]. $N = 30$ gives power ≥ 0.999 against the pilot's full framing-effect magnitude on the pass-rate metric and ≥ 0.80 down to ∼60% attenuation, but is explicitly underpowered for a CCEI-scale effect of the pilot's magnitude at 1.5B (minimum detectable ≈ 0.11), which we report rather than paper over. Every replicate's Bronars power [Bronars, 1987] is recorded; all 150 draws held power ≥ 0.998.

4.3 The discard-selection problem, and its correction as a stated contribution

A pilot run found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required output-format contract and were silently discarded under the pilot's own naive handling — standard practice, and one this paper measures the consequences of directly rather than assumes. We treat this as a claim about instrument validity in its own right, not a footnote to the dose–response result, true regardless of whether projection helps, hurts, or does nothing downstream. The main experiment implements a capped retry protocol: a session failing the $\geq 20/25$-valid-rounds threshold is retried up to three total attempts with a fresh seed offset, every attempt recorded; a slot still failing after three attempts is a residual discard, excluded from CCEI/GARP but retained for audit.

Table 2 breaks every reciprocal-framing session down by attempt outcome — kept on the first attempt, rescued by a later retry, or still discarded after three attempts — for both models. At 1.5B, retry-rescued sessions ($n = 7$, recovered on attempt 2 or 3) are not a close stand-in for first-attempt sessions ($n = 17$): they show *higher* mean CCEI (0.965 vs. 0.932) but a substantially *lower* GARP pass rate (14% vs. 47%). The sessions that never produce a usable trace even after three attempts ($n = 6$, three with zero valid rounds recorded at all) show the lowest CCEI among the three groups on the handful that can be scored (0.882, $n = 3$) — consistent with, though not conclusive given the small residual sample, some of the same selection concern the pilot's naive handling raised persisting at a smaller scale even after correction; the retry protocol reduces but does not eliminate it. At 3B, by contrast, the first-attempt and retry-rescued groups sit much closer together on CCEI (0.974 vs. 0.962) than the 1.5B groups do, and the small residual-discard group ($n = 2$) is trivially GARP-consistent; the selection concern that motivates this section shows up more sharply at 1.5B than at 3B, though the 3B residual group is too small to draw a firm conclusion from. We report first-attempt and residual post-retry discard rates as a first-class per-condition outcome for every cell.

5 Results

187 attempt-records were collected across 150 replicate slots (2 models × up to 3 conditions × 30 replicates); 142 slots kept a usable trace, 8 were residual discards after three attempts. Every cell reached its full 30 replicates.

Table 2: Discard-outcome breakdown for all 60 reciprocal-framing sessions (30 per model), by attempt group: kept on the first attempt, rescued by a later retry, or still discarded after three attempts (§4.3). GARP pass and CCEI use the same methodology as Table 3; mean dose (L_1) is averaged over the full group, with 0 for any already-GARP-consistent trace. *qwen residual discard: 3 of the 6 slots returned zero valid rounds on every attempt (no p/x data at all); the reported CCEI, GARP pass rate, and dose for this row are computed over the remaining 3 slots only.

Model	Attempt group	n	mean CCEI	GARP pass	mean dose (L_1)
llama3.2:3b	first-attempt success	21	0.9742	0.4286	5.086
llama3.2:3b	retry-rescued	7	0.9624	0.2857	9.839
llama3.2:3b	residual discard	2	1.0000	1.0000	0.000
qwen2.5:1.5b	first-attempt success	17	0.9315	0.4706	13.847
qwen2.5:1.5b	retry-rescued	7	0.9651	0.1429	7.600
qwen2.5:1.5b	residual discard	6	0.8821*	0.6667*	24.196*

Table 3: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are computed on the subset of kept traces that violate GARP.

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multiturn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

5.1 The dose–response relationship, and the control that overturns it (C1)

Across all 85 GARP-violating traces with a computed, independently-verified projection (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$), Pearson $r = 0.821$ ($p < 0.0001$), mean Δpayoff $+0.0091$ (one-sample $t = 5.41$, $p < 0.00001$), with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. Figure 2 shows the relationship split by model on shared axes. **Both scales show a significant, positive relationship, and it is stronger at the headroom model**: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B versus $\rho = 0.614$ ($p = 0.0011$, $n = 25$) at 3B. We tested the mechanical confound that larger-dose traces might simply start from worse raw payoff and therefore have more room to improve. The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain the relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$), and an OLS regression of Δpayoff on dose and raw payoff jointly finds dose highly significant ($t = 11.4$, $p < 10^{-16}$) with raw payoff's own marginal contribution not distinguishable from zero ($p = 0.18$) once dose is included. Full detail in Appendix A.

This is where a paper that stopped at the raw correlation would end, and it is not where this one ends. The confound just tested is a confound on the *severity* of the raw violation, not on the specific *geometry* of the payoff function — it says nothing about whether restoring GARP-consistency *specifically*, rather than merely displacing choices toward some interior point, is what buys the payoff gain above. §3.4 builds the control that closes this gap: a GARP-blind null operator, spending the identical L_1 displacement budget as the real repair, that does nothing but shrink every bundle toward the exogenous payoff's own fixed optimum.

Experiment 1 (original payoff). The null operator outperforms the real repair by roughly $2.4\times$: mean $\Delta\text{payoff}_{\text{null}} = 0.0220$ versus mean $\Delta\text{payoff}_{\text{real}} = 0.0091$, Wilcoxon signed-rank $p = 3.98 \times 10^{-10}$, with the null winning 69 of 85 traces (81%). This holds independently at both models. The partial correlation of dose against $\Delta\text{payoff}_{\text{real}}$, controlling for the null's own mechanically-predicted gain, remains positive and significant ($r = 0.574$, $p = 1.13 \times 10^{-8}$) — dose retains real information beyond what the null predicts — but the paired comparison is unambiguous: at the identical displacement budget, an operator with no knowledge of GARP buys more than twice the payoff the real repair does.

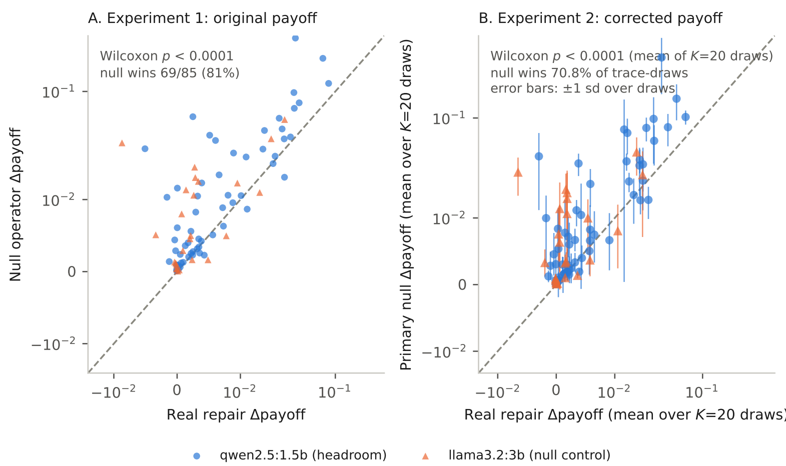

Figure 2: Real repair's Δpayoff (x-axis) against the distance-matched null operator's Δpayoff (y-axis), one point per GARP-violating trace ($n = 85$), marker shape/color by model (circle, blue: qwen2.5:1.5b; triangle, orange: llama3.2:3b). The dashed line is $y = x$; points above it are traces where the null operator's payoff gain exceeded the real repair's. Both axes use a symmetric-log scale (linear within ± 0.01, logarithmic beyond it) so that one far-larger-dose trace does not compress the other 84 points into one corner. **(A)** Experiment 1: the original, fixed-target Cobb–Douglas payoff (§3.3). **(B)** Experiment 2: the corrected, per-trace random-target payoff (§3.4), each trace's Δpayoff averaged over its $K = 20$ independent draws of the random target α_s, with vertical error bars showing ± 1 SD of the null operator's Δpayoff across those draws; the primary (information-fair) null construction is plotted, not the oracle null. Annotated Wilcoxon signed-rank p-values and null win rates are the statistics reported in §5.1.

Experiment 2 (corrected payoff). The original payoff's exploit is a single fixed target ($s = 0.5$) identical across every trace; §3.4 removes this by drawing an independent Cobb–Douglas weight $\alpha_s \sim \mathrm{Uniform}(0.05, 0.95)$ per trace, repeated across $K = 20$ independent draws for robustness. The qualitative result is unchanged: the primary, information-fair null still outperforms the real repair (mean $\Delta\text{payoff}_{\text{null}} = 0.02167$ versus mean $\Delta\text{payoff}_{\text{real}} = 0.00723$, Wilcoxon $p = 1.24 \times 10^{-5}$, win rate 70.8%), significant in 20 of 20 draws. An oracle null privileged to know the true per-trace target does better still (mean $\Delta\text{payoff}_{\text{null}} = 0.02224$, $p = 1.35 \times 10^{-9}$, win rate 80.4%, significant in 20/20 draws), reported for completeness rather than as the primary comparison. The partial correlation of dose with $\Delta\text{payoff}_{\text{real}}$, controlling for the null, attenuates from Experiment 1's $r = 0.574$ to a mean $r \approx 0.37$ across the 20 draws, significant in 14–15 of 20 — a real but more modest signal, not distinguishable from the draw-to-draw noise the 20 draws themselves already show.

No third payoff was attempted, and this is a stated scope decision, not a resource constraint. Two independently designed, independently motivated payoffs — one the paper's original, one built specifically to remove the first one's known exploitable structure — converging on the same negative paired-comparison result is the finding. Redesigning the payoff a third time only because the second attempt also came back negative would be payoff-shopping: iterating the yardstick until GARP-restoration happens to look good, exactly the researcher-degree-of-freedom problem an exogenous payoff was introduced to rule out.

A partial mechanistic note. Trace extremity ($1 -$ raw payoff under the original fixed payoff, a stable measure of the raw bundles independent of which payoff later scores them) predicts the null's advantage strongly in both experiments (Experiment 1: Pearson $r = 0.679$; Experiment 2 primary null: mean $r = 0.631$ across 20 draws), more strongly than it predicts the real repair's own gain in either. This pattern alone does not cleanly separate "centering" from a more general "moving an extreme trace toward the interior helps, regardless of exactly where" account, since Experiment 2's primary null is mechanically the same fixed-center operator as Experiment 1's. Experiment 2's oracle null, which targets a genuinely different, per-trace-varying point, shows essentially the same extremity-advantage (mean $r = 0.624$) — read as partial, not conclusive, support for an "escape a bad start" mechanism rather than pure centering, offered as a discussion-level hypothesis rather than

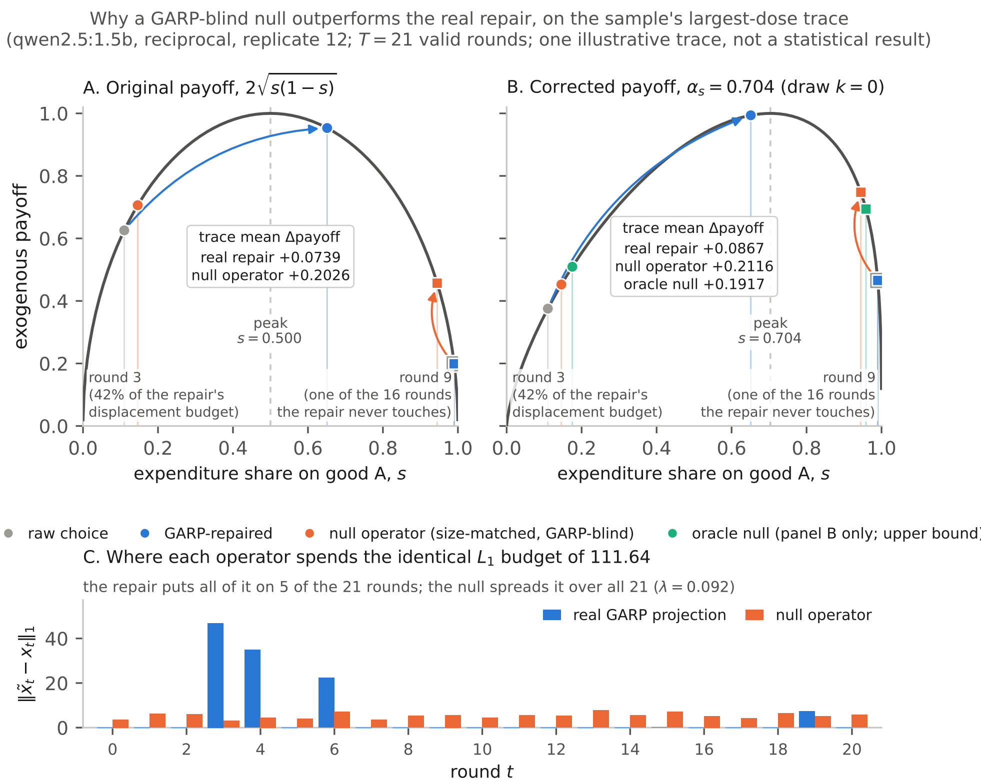

Figure 3: Why the size-matched, GARP-blind null outperforms the real repair, shown on one trace rather than argued from aggregate statistics. With $K = 2$ and budget exhaustion the payoff is exactly a function of the expenditure share s on good A: $2\sqrt{s(1-s)}$ under the original design (panel A, peak fixed at $s = 0.5$ for every trace) and $s^{\alpha_s}(1-s)^{1-\alpha_s}/[\alpha_s^{\alpha_s}(1-\alpha_s)^{1-\alpha_s}]$ under the corrected one (panel B, peak at this trace's own draw $\alpha_s = 0.704$, draw $k = 0$ of the 20). The trace is qwen2.5:1.5b, reciprocal, replicate 12 — the largest-dose violation in the sample ($L_1 = 111.64$, $T = 21$ valid rounds), hand-audited in Appendix A; every plotted quantity is recomputed by re-solving the projection MILP. Panel C gives the mechanism: the minimal-perturbation repair spends its whole displacement budget on 5 of the 21 rounds (42% on round 3 alone) and leaves the other 16 untouched, where the null spreads the identical budget over all 21. *On the rounds it does touch the repair beats the null* — at round 3 it moves the share $0.11 \to 0.65$, payoff $0.626 \to 0.953$ against the null's 0.706 — and loses on the trace average only because the 16 untouched rounds sit at a near-corner $s = 0.99$ and stay there (payoff 0.199) while the null lifts each of them (to 0.457). Panel B shows this surviving the corrected payoff: even aiming at the wrong point (0.5, not 0.704), the null still gains more. **One trace and two of its rounds, chosen for legibility: an illustration, not a statistical result** — the evidence is the 85-trace paired comparison in this section and Appendix A.

a load-bearing result with its own significance test. Figure 3 illustrates the same idea concretely on a single trace: the rounds that generate the null's advantage there are precisely the extreme-share rounds the real repair leaves untouched.

C1 as originally proposed — that restoring GARP-consistency specifically, beyond mere displacement toward the interior of the budget line, buys more exogenous payoff — is not supported under either payoff design. The raw dose–response relationship reported at the top of this section is real; it is not evidence for the specific mechanism C1 claims.

5.2 The reciprocal-framing manipulation: survivorship, and a correction that helps but does not fully resolve it

At 3B, reciprocal framing replicates the pilot's finding on an independently-drawn budget-set design: GARP pass rate collapses from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$ (survives Benjamini–Hochberg correction across our full test family, $p_{\mathrm{BH}} = 0.011$, but not the more conservative Holm correction, $p_{\mathrm{Holm}} = 0.062$); CCEI moves much less and reaches only borderline significance (0.9900 vs. 0.9713, t-test $p = 0.0508$) — the same near-1 compression the pilot flagged, and a further case in this paper of the pattern that GARP pass rate is the sensitive instrument for framing/format disruption while CCEI understates it.

At 1.5B, the manipulation this study was designed around does *not* survive proper correction for discard-selection. First-attempt discard under reciprocal framing at 1.5B was 43.3% (13/30), falling to a residual 20.0% (6/30) after the retry protocol — smaller than the pilot's unretried 52%, plausibly because independent per-replicate budget sets less often reproduce one unusually hard draw, but the pattern is unchanged: reciprocal framing produces far more discards than any other condition, including the *other* 1.5B manipulation (multiturn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing, with confidence intervals overlapping almost entirely. This is no detectable CCEI shift, not a confirmation of the pilot's own naive point estimate: the pilot's $+0.0169$ ($p = 0.66$, itself not significant) and the main experiment's -0.0109 ($p = 0.73$) are two statistically indistinguishable nulls, and reading the difference in their signs as a reversal of a real effect overstates what either number supports. Table 2 in §4.3 gives the finer-grained per-attempt-outcome breakdown that these top-line pilot and main-experiment numbers average over.

5.3 The multiturn/format effect: a large effect in the opposite direction from the literature

Splitting the 25 rounds into separate sequential calls collapses the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI barely moves (0.9522 vs. 0.9540, $p = 0.91$; Figure 4) — a larger drop in percentage points than reciprocal framing produced at 1.5B, with *zero discards on either arm* so no selection-bias caveat applies. Corrected for multiple comparisons across our full test family, this result survives Benjamini–Hochberg correction ($p_{\mathrm{BH}} = 0.0094$) but not the more conservative Holm correction ($p_{\mathrm{Holm}} = 0.058$); we report both rather than the raw p-value alone. Together with the 3B reciprocal-framing result (§5.2), this is a further case of the same pattern. Against the literature this arm was designed against [Wang et al., 2025a]: in the same budget-allocation domain, on the same model family at a larger scale (Qwen2.5-7B), single-turn elicitation was found to be *more* coherence-degrading than multi-turn; here, at 1.5B, the reverse holds — multi-turn elicitation collapses GARP pass rate relative to our single-turn baseline. We do not have an explanation for the reversal and do not offer one; we report the disagreement as a finding in its own right, not a confirmation of the literature's own strongest lever.

6 Limitations

CCEI is noisy at 1.5B, and the paper leads with GARP pass rate because of it, not by preference. The 1.5B model's within-condition CCEI noise (baseline within-subject coefficient of variation 14.8%) is comparable to published human test–retest noise on the same index [Nitsch et al., 2022], and $N = 30$ is explicitly underpowered for a CCEI-scale effect of the pilot's own magnitude at this scale (the minimum reliably detectable effect at this N is roughly $2.5\times$ the pilot's magnitude). A CCEI-power-matched design would need $N \approx 111$ to 161 depending on the assumed baseline variance; that scale-up was deferred pending this run's results and was not executed. We treat this as an open design tradeoff, not a result masked by a metric choice: the pass-rate metric is adequately powered throughout and every CCEI result we report is accompanied by its own confidence interval rather than presented as equally reliable.

Single run per condition per model at $N = 30$. Every (model, condition) cell in this paper is one run of the stated protocol; we do not have repeated full runs of the entire pipeline to report a between-run variance on the headline dose–response statistic itself, only the within-run replicate variance captured by the reported confidence intervals and p-values.

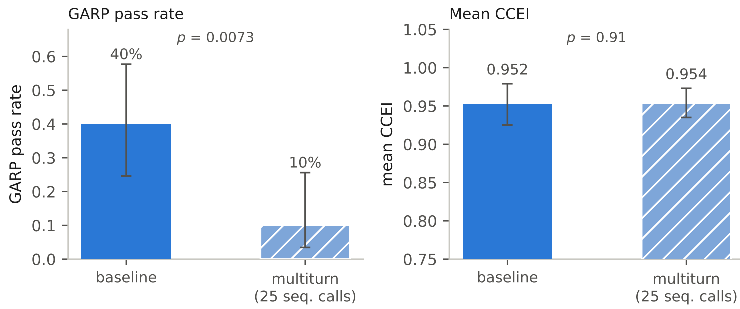

Figure 4: 1.5B (qwen2.5:1.5b), baseline vs. multiturn elicitation format (25 sequential calls), $n = 30$ per condition, zero discards in either arm — unlike the reciprocal-framing condition (Table 2). Left: GARP pass rate, with 95% Wilson score confidence intervals on each bar (baseline 12/30 = 40% [95% CI 24.6–57.7%]; multiturn 3/30 = 10% [95% CI 3.5–25.6%]; Pearson χ^2 test, $p = 0.0073$). Right: mean CCEI, with 95% confidence intervals computed from the reported mean, SD, and n via the t-distribution ($df = 29$) (baseline 0.952 [95% CI 0.925–0.979]; multiturn 0.954 [95% CI 0.935–0.973]; two-sample t-test, $p = 0.91$). The second bar in each panel is additionally hatched so the two conditions remain distinguishable without color.

Two model families, one scale point each. The headroom/null-control design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* model scale: a pilot found headroom and measurement reliability do not coexist anywhere in the range piloted, so the design targets the scale where headroom exists rather than blending a noisy-but-real candidate effect with a clean-but-absent one across a wider range (a third family was excluded after the multi-model-residency failure noted in §4; extending the roster is a documented follow-up, not a silent omission).

Two fixed exogenous payoffs, not three. We report two independently designed payoffs (§3.3, §3.4) because the first one's single, globally-identical optimum turned out to be exploitable by a GARP-blind operator; the second removes that specific property and reproduces the same verdict. We do not report a third payoff. A third, run only because the second also came back negative, would be payoff-shopping rather than a robustness check, and §5.1 says so rather than quietly running one until the result flips.

An adverse prior the paper now joins rather than argues against. Three independently published repair attempts moved the wrong way (§2). Our own reciprocal-framing manipulation at 1.5B (§5.2) is a fourth negative in the narrower sense that the manipulation intended to induce measurable coherence variation did not do so once properly measured. The null-operator result of §5.1 is a fifth: under two independently designed exogenous payoffs, restoring GARP-consistency specifically does not outperform a GARP-blind operator matched on displacement. We do not read this as evidence that the coherence–competence relationship is negative in general — C2 remains an open identification question, not a settled sign — only that this paper's own attempt to find a positive relationship, net of mere displacement, did not succeed.

Broader impacts. This paper's central result argues directly against imposing GARP-restoration on deployed AI economic agents as a default decision-quality intervention: under two independently designed exogenous payoffs, a GARP-blind operator that merely displaces choices toward an interior point outperforms the real repair (§5.1). A practitioner who deploys coherence-repair expecting a decision-quality gain, on the strength of a raw dose–response correlation alone, would be adopting exactly the confound this paper's control was built to catch. The reciprocal-framing manipulation this study was designed around similarly turned out, once corrected, to have no measurable effect on the coherence it was thought to disrupt (§5.2). The paper's contribution is a measurement design for asking whether repair helps in a given setting — and, in the setting it tests, a documented instance where it does not — not a recommendation to deploy it universally, and not a demonstration that no coherence-repair method could ever help downstream decisions. Separately, the discard-selection finding (§4.3) generalizes beyond this paper: any revealed-preference or consistency instrument that silently drops unparseable or disruption-caused failures risks systematically underestimating exactly the manipulations that matter most, which bears on how AI economic-behavior audits are designed and reported more broadly. We see no elevated misuse risk specific to this work: it uses only synthetic budget-allocation tasks on publicly available open-weight models, releases no new model or dataset, and does not involve human subjects.

7 Conclusion

This paper set out to measure whether restoring GARP-consistency in an LLM agent's own realized choices buys more exogenous payoff than the mechanical fact of displacement alone. Applying Demuynck and Rehbeck [2023]'s minimal-quantity-error GARP repair post hoc to a frozen agent, we find a real, precisely estimated, monotone raw dose–response relationship between repair size and payoff gain. But a distance-matched, GARP-blind null operator — absent from every prior axiom-enforcement result we are aware of — outperforms the real repair on that same payoff, significantly, at both model scales; a second, independently designed payoff built specifically to remove the first one's exploitable structure reproduces the same verdict. C1, as originally proposed, is not supported: the answer this paper's control was built to give is no. A partial mechanistic finding — trace extremity predicts the null's advantage in both experiments — offers a candidate explanation without fully resolving it. What survives is the paper's identification argument itself (C2): the coherence–competence relationship remains open, not settled in either direction, and this is the first attempt we know of to measure it without confounding coherence with capacity, a preference-judgment outcome, or mere displacement magnitude. Two further, unpredicted findings — a purpose-built framing manipulation's apparent 1.5B effect was a discard-selection artifact rather than a real coherence shift (C3), and an elicitation-format manipulation produces a large GARP-pass-rate collapse in the opposite direction from an independently published finding in the same domain and model family at a larger scale — are reported in their own right. A well-controlled negative, an open identification question, and an instrument-validity finding are exactly the kind of result this literature should surface by default rather than paper over.

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A Payoff exogeneity audit

Full detail of the four-part adversarial audit summarized in §3.3.

(1) No shared derived quantity beyond the exogenous (p, I). The payoff implementation has no dependency on the projection implementation; the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The one near-miss is that the projection's feasibility incumbent is a Cobb–Douglas demand *share-fitted to the agent's own observed data* — a plausible leakage channel on first read — but this quantity is never passed to the underlying MILP solver as an actual incumbent (the solver used has no such interface); it is used only to log an upper-bound sanity ceiling on the reported distance, never to steer the solution.

(2) The payoff's optimum is independent of revealed preference, checked empirically as well as by construction. Comparing raw (pre-repair) payoff between the 57 traces that were already GARP-consistent and the 85 that were not: mean 0.7872 vs. 0.7899, Welch $t = -0.07$, $p = 0.94$. GARP-consistency status alone does not predict payoff, ruling out a population-level mechanical link between coherence and payoff score.

(3) Hand-derived mechanism check. Five traces spanning the full dose range (0.17 to 111.64) were inspected directly, comparing the direction of the actual projection movement $(\tilde{x} - x)$ against the direction toward the exogenous optimum $(x^* - x)$ via cosine similarity. Similarities were low (0.16–0.31) on the four traces where payoff improved after repair, and *negative* (-0.35) on the single trace where payoff worsened — the sign of the outcome tracks a geometric quantity the projection objective never references, evidence against the mechanism being a disguised optimization toward the payoff target.

(4) The severity-confound check. Larger-dose traces do start from worse raw payoff (Pearson $r = -0.41$, $p < 0.0001$), and worse-starting traces do have more room to improve ($r = -0.41$ between raw payoff and Δpayoff) — a genuine ceiling-effect confound. It does not explain the dose–response relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$, attenuated only slightly from the unconditional $r = 0.821$), and in a joint OLS regression the dose coefficient is highly significant ($t = 11.43$, $p < 10^{-16}$) while raw payoff's own coefficient is not ($t = -1.36$, $p = 0.18$).

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   Justification: The abstract and §1 state exactly what is claimed (a measured dose–response relationship, an instrument-validity finding, and a replication of an independently published format effect) and what is not (novelty of repair, of projection, or of the sign of the effect); §6 states the design's scope limits explicitly.

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   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §6 (Limitations) discusses CCEI's noise and power at 1.5B, the single-run-per-condition design, the two-model-family scope, the single fixed payoff weighting, and the adverse prior from three prior negative results.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
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Justification: The paper's contribution is a computational method (a MILP formulation for GARP-repair) building on a cited external representation theorem [Demuynck and Rehbeck, 2023]; it does not state or prove new theorems of its own.

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